

Wellington Santos

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with **8+** years across sports commerce, e-commerce retail, and healthcare IT, designing and deploying machine learning models for production systems. Proficient in Python, TensorFlow, and PyTorch with expertise in data pipelines, model optimization, and MLOps infrastructure. Experienced building scalable AI solutions using cloud platforms and modern ML frameworks.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Generative AI: LangChain, OpenAI API, Anthropic API, LlamaIndex, Hugging Face, LangGraph, Ollama

Data & ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Spark, Airflow, MLflow, Weights & Biases, dbt

Backend: FastAPI, Django, Node.js, Express, Spring Boot, RabbitMQ, Kafka

Database: PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch

Testing: Pytest, Jest, unittest, Pytest-cov, Selenium

Cloud & DevOps: AWS, GCP, Azure, SageMaker, Vertex AI, Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, CloudWatch

Others: RESTful API, GraphQL, Git, GitHub, Jira, Confluence, Agile, Scrum

EXPERIENCE

Fanatics

Oct 2023–Present

Senior AI/ML Engineer

- Led the development of a real-time recommendation engine for a sports fan engagement platform using **Python** and **PyTorch**, deploying collaborative filtering models that increased average daily active user engagement by **28%**.
- Spearheaded the design and implementation of a customer churn prediction model using **TensorFlow** and **scikit-learn** to identify at-risk users across collectibles and sports betting segments, enabling targeted retention campaigns.
- Played a key role in establishing MLOps infrastructure by integrating **MLflow** for model versioning and experiment tracking, standardizing model deployment workflows across three production services.
- Implemented a feature engineering pipeline using **Pandas** and **NumPy** that automated extraction of user behavior signals from Kafka event streams, improving model training efficiency by **35%**.
- Deployed a classification model for content personalization to production on **AWS SageMaker**, building inference APIs in **FastAPI** that handled 50k+ daily prediction requests with sub-100ms latency.

- Collaborated with the data engineering team to optimize training data refresh cycles, reducing model retraining time from 8 hours to 2 hours through **Spark** distributed processing.
- Conducted A/B tests on three competing recommendation algorithms by validating holdout sets and monitoring business KPIs, presenting findings to product and leadership teams to guide model selection.
- Mentored two junior engineers on best practices for model evaluation, feature engineering, and production monitoring, establishing code review standards that reduced model-related production incidents by **40%**.
- Implemented automated model drift detection using **Weights & Biases** and custom monitoring dashboards, enabling the team to catch model degradation within hours instead of days.
- Built a batch prediction pipeline using **Kubernetes** and **Docker** to score millions of user profiles daily, integrating outputs with the marketing platform via **PostgreSQL** and **RESTful API** endpoints.
- Experimented with LLM-based content recommendation using **LangChain** and the **OpenAI API**, prototyping personalized product descriptions for the collectibles marketplace to assess commercial viability.

Walmart

Sep 2022–Oct 2023

Senior ML Engineer

- Spearheaded the development of a demand forecasting model for e-commerce inventory management using **TensorFlow** LSTM networks and historical sales data, reducing excess inventory carrying costs by **22%** across 15 distribution centers.
- Built end-to-end training and inference pipelines using **Airflow** and **Python** that processed terabytes of transaction data weekly, ensuring models stayed current with seasonal and promotional demand patterns.
- Collaborated with the data platform team to design a feature store using **PostgreSQL** and **Redis**, enabling rapid experimentation with new features across five concurrent model development projects.
- Deployed fraud detection classifiers to production on **AWS** Lambda behind **RESTful API** gateways, processing real-time payment transactions and reducing transaction fraud by **18%** within the Walmart Pay fintech division.
- Optimized a computer vision model for in-store shelf monitoring using **PyTorch** quantization and knowledge distillation techniques, reducing model inference latency from 800ms to 280ms on edge devices.
- Established comprehensive model monitoring using **Prometheus** and **Grafana**, tracking prediction accuracy, latency, and data drift across production models to alert engineers to retraining needs.
- Led cross-functional workshops on ML best practices and responsible AI principles with engineers, product managers, and compliance teams, establishing governance standards for model transparency.
- Experimented with retrieval-augmented generation using **LlamaIndex** and **Anthropic Claude API** to build an internal documentation assistant, proving feasibility for rollout to store associates.
- Mentored a data scientist on software engineering discipline, conducting code reviews and establishing version control workflows that reduced deployment bugs by **33%**.

E-POT

Jun 2018–July 2022

ML/Full Stack Engineer

- Built machine learning solutions for healthcare IT systems, developing predictive models using **Python**, **scikit-learn**, and **TensorFlow** to forecast patient readmission risk within 30 days, achieving **89%** recall.
- Designed and deployed a natural language processing pipeline using **Hugging Face** transformers to extract clinical insights from unstructured medical notes, enabling automated triage recommendations.
- Implemented a medical imaging classification system using **PyTorch** convolutional neural networks for radiology reports, training on 50k+ labeled images and integrating results into hospital dashboards via **FastAPI**.
- Collaborated with clinical data engineers to build ETL workflows in **Airflow** and **Spark**, processing multi-source patient data streams and maintaining feature consistency across model environments.
- Developed a time-series anomaly detection model using **PyTorch** LSTMs to monitor patient vital signs in real-time, triggering alerts when physiological parameters deviated from learned baselines.
- Maintained feature engineering code in **Git** repositories with comprehensive test coverage using **Pytest**, establishing CI/CD automation in **GitHub Actions** for reproducible model training jobs.
- Experimented with transfer learning approaches for medical imaging across multiple hospital systems with limited labeled data, reducing training data requirements by **60%** while maintaining diagnostic accuracy.

EDUCATION

University of Tokyo

Apr 2014–Mar 2018

Bachelor of Science in Computer Science